



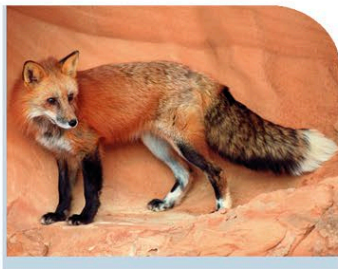
CLASSIFYING INSECTS

About 20,000 species of butterfly have now been described. In order to identify them or recognize new species, museums hold vast reference collections, like this one, for comparative purposes.



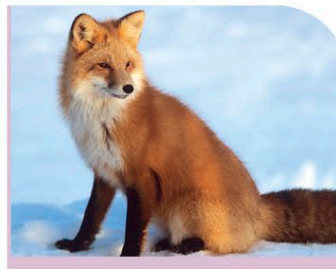
ORDER *Carnivora*

Orders are the next level down in Linnaeus's hierarchy, and contain one or more families. The *Carnivora* have modified cheek teeth (carnassials) and large canine teeth specialized for biting and shearing.



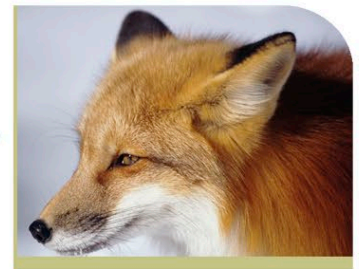
FAMILY *Canidae*

A subdivision of an order, a family is made up of genera and the species within them. The family *Canidae* has 35 living species, all with non-retractile claws and two fused wrist bones. All but one species have long, bushy tails.



GENUS *Vulpes*

A term first used by Aristotle in Ancient Greece, a genus identifies subdivisions of a family. *Vulpes* is a genus within the family *Canidae*. All foxes have upright, triangular ears and a long, narrow, pointed snout.



SPECIES *Vulpes vulpes*

The basic unit of taxonomy, species are populations of similar animals that breed only with one another. *Vulpes vulpes*, known for its bright red fur, breeds only with other red foxes.

ANIMAL, VEGETABLE, OR MINERAL



Linnaeus divided the natural world into three kingdoms – animals, plants, and minerals. He then devised a hierarchical system of classification based on class, order, family, genus, and species, and established the convention of giving species binomial Latin names.

CARL LINNAEUS, 1707–1778

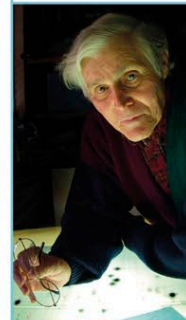
A NEW KINGDOM



Historically organisms were classed as being either animals or plants, but in 1866 Ernst Haeckel argued that microscopic organisms formed a separate group, which he called *Protista* (now *Protoctista*). There were now three kingdoms of life – *Animalia*, *Plantae*, and *Protista*.

ERNST HAECKEL, 1834–1919

INTRODUCING THE ARCHAEA



Recognized by Carl Woese and George Fox in 1977, the *Archaea* are microscopic organisms that live in very extreme environments. Initially grouped with the *Bacteria*, their DNA turned out to be so unique that a new three-domain taxonomic system was introduced.

CARL WOESE, BORN 1928